Software Requirements Specification

for

"InnerBloom :Mental Wellbeing Platform "

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1. Introduction

1.1 Purpose

State the primary objectives for this document by outlining the full suite of functional and operational specifications for the **InnerBloom** application, which may also be identified by its internal **INNERBLOOM** module designation. This section serves to bridge fundamental organizational strategies with formal software engineering design constraints.

1.2 Document Conventions

This document adheres to the strict standard structural patterns defined by IEEE 830. To maintain traceability and clear engineering execution, every functional requirement statement is uniquely indexed with an alphanumeric sequence tag (REQ-X). Detailed requirements automatically inherit the top-level priority matrix of their parent subsection feature.

1.3 Intended Audience and Reading Suggestions

This engineering resource is explicitly targeted at system architects, database administrators, frontend developers building interface logic, and validation/QA engineers mapping operational use-cases. It is recommended to read the **Section 2 (Overall Description)** for a structural overview before analyzing the core features and routing behaviors mapped in **Section 4 (System Features)**.

1.4 Product Scope

InnerBloom serves as a premium, secure, and user-centric mental health platform engineered to bridge the gap between patients seeking psychological evaluation and medical practitioners managing consultation workflows. The system incorporates data-driven mental wellness assessment tools, interactive self-tracking dashboards, gated educational resources, and a stringent credential validation portal for medical professionals to guarantee premium care quality.

2. Overall Description

2.1 Product Perspective

InnerBloom is an autonomous, cloud-coexistent software architecture acting as a multi-tier web application platform. It interfaces securely with relational database engines for user profile persistence, file storage nodes for binary medical documentation, and a centralized system administration portal for credential justification.

2.2 Product Functions

- **User Identity Management:** Automated registration, string-validated input processing, and error-handled session authentication.
- **Session Lifecycle Routing:** Direct automated interface routing logic shifting system views to target consoles upon successful form validation.
- **Metric Logs & Content Distribution:** Persistence layers for tracking real-time daily meditation times alongside premium streaming filters for therapeutic video logs, informational assets, and magazine publications.
- **Clinician Asset Onboarding:** A rigorous, multi-checkbox credential registration interface designed for document attachment and sequential state tracking.
- **Clinical Agenda Portals:** Analytical workspace layouts summarizing daily patient consultation volume and auxiliary calendar shifts.
- **Diagnostic Point Matrix Engine:** Algorithmic logic processing questionnaire selections into point-based tiered diagnostic evaluations.

2.3 User Classes and Characteristics

Unregistered Guest: Limited to baseline organizational marketing landing scopes, system sign-up pages, and preliminary test forms.
Registered Subscribed Patient: Accesses active consultation appointments, daily time trackers, and locked multimedia educational asset logs.
Authorized Medical Professional (Doctor): Accesses dedicated clinical workspaces containing task lists, client tracking tools, and active agenda managers.
System Administrator: Oversees cross-system verification workflows, executes doctor document justification reviews, and manages gated publication entities.

2.5 Design and Implementation Constraints

UI Symmetrical Canvas Parity: Front-end developments must strictly replicate a clean grid layout and borderless glassmorphism card containers.
Standard Width Framework: Every digital view asset must be rendered within a standardized horizontal aspect ratio matrix to ensure visual stabilization and eliminate layout shifting across viewport changes.

2.6 User Documentation

User Manual (PDF/Print): A comprehensive guide covering installation, configuration, and usage instructions.
Online Help System: Context-sensitive help integrated within the application, accessible via the Help menu or F1 key.
Quick Start Guide: A concise tutorial highlighting the most common tasks and workflows.
Video Tutorials: Short instructional videos demonstrating key features and processes.
FAQs and Knowledge Base: A searchable repository of frequently asked questions and troubleshooting tips.

2.7 Assumptions and Dependencies

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3. External Interface Requirements

3.1 User Interfaces

The software will provide intuitive and accessible user interfaces designed for different user roles (patients, doctors, and administrators).

Logical Characteristics:

- **Login/Signup Screens:** Clean forms with validation, password masking, and error messages displayed in a consistent format.
- **User Dashboard:** Displays assessment results, appointment schedules, subscription status, and chatbot access.
- **Doctor Dashboard:** Provides patient enrollment details, appointment management, and secure communication tools.
- **Chatbot Interface:** Simple text box with send button, supporting real-time responses.
- **Assessment Module:** Interactive questionnaire with progress indicators and clear navigation buttons (Next, Back, Submit).
- **Error Handling:** All error messages will be displayed in modal pop-ups or inline alerts with consistent styling (red text, warning icon).
- **Help Button:** A standard help icon will appear on every screen, linking to online documentation.
- **GUI Standards:** Interfaces will follow responsive design principles (HTML5, CSS3) and use consistent color schemes and typography across all modules.
- **Keyboard Shortcuts:** Common actions such as Ctrl+S for save and Esc for cancel will be supported.

Components Requiring UI:

- Registration/Login
- User Dashboard
- Doctor Enrollment
- Appointment Booking
- Assessment Results

3.2 Hardware Interfaces

The system is primarily web-based and will interact with hardware components indirectly through supported devices.

Logical and Physical Characteristics:

- **Supported Devices:**
  - Desktop and laptop computers (Windows, macOS, Linux)
  - Mobile devices (Android, iOS smartphones and tablets)
- **Data and Control Interactions:**
  - User input via keyboard, mouse, and touchscreen.
  - File uploads (e.g., doctor credentials, medical documents) through standard browser file dialogs.
- **Communication Protocols:**
  - HTTPS for secure client-server communication.
  - SSL/TLS encryption for data transfer.
- **Peripheral Support:**
  - Email notifications via SMTP servers.
  - Optional integration with webcam/microphone for tele-consultation modules (future scope).

3.3 Software Interfaces

The system will interact with several software components to deliver its functionality:

Databases:

- **MySQL 8.x** – Primary relational database for storing user accounts, doctor enrollment data, appointments, and assessment results.
- **NoSQL (optional Firebase Firestore)** – For real-time data storage in chatbot and notification modules.

Operating Systems:

- Compatible with **Windows 10/11**, **Linux (Ubuntu 20.04+)**, and **macOS Monterey+** for server deployment.
- Client access through modern browsers (Chrome, Edge, Firefox, Safari).

Libraries and Tools:

- **PHP 8.x** – Backend scripting language for REST API endpoints.
- **JavaScript (ES6+)** – Frontend logic and API calls.
- **CSS3/HTML5** – User interface styling and structure.

- **FontAwesome 6.x** – Icon library for UI consistency.
- **Bootstrap 5.x** – Responsive design framework.
- **Encryption Libraries (OpenSSL, PHP password_hash)** – For secure password storage and SSL/TLS communication.

Integrated Components:

- **REST API:** For login form more secure, reusable, and professional by cleanly connecting the UI with the backend authentication system.
- **Payment Gateway (SSLCOMMERZ)** – For subscription and doctor consultation fees.

Data Flow:

- **Incoming Data:** User registration details, login credentials, doctor enrollment forms, assessment responses, chatbot queries.
- **Outgoing Data:** JSON responses for API calls, appointment confirmations, subscription status, chatbot replies, error messages.

Communication Nature:

- RESTful APIs using **JSON** format for request/response.
- Secure communication via **HTTPS**.
- Shared data across modules includes **user profile info, appointment schedules, subscription status, and assessment scores**.

3.4 Communications Interfaces

The system requires robust communication interfaces to ensure secure and reliable data exchange:

Web Browser Communication:

- All client interactions occur via modern browsers using **HTTPS protocol**.
- Standard message formatting in **JSON** for API responses.

Standards:

- **REST API** for structured communication between frontend and backend.
- **SSL/TLS** for encryption.
- **UTF-8** encoding for all text data.

Security and Performance:

- All sensitive data (passwords, medical records) encrypted before transmission.
- Communication rate optimized for low latency in chatbot responses.

- Synchronization mechanisms ensure consistency between user dashboard and doctor dashboard.

4. System Features

4.1 New User Registration Form

4.1.1 Description and Priority

Enables a generic guest user to initialize an authenticated account status within the relational database profile repository. (Priority: High)

4.1.2 Stimulus/Response Sequences

- **User Action:** Clicks the "Sign Up" trigger button on the structural sidebar panel.
- **System Action:** Generates a blank, borderless card-layered data schema collection form.
- **User Action:** Inputs coordinates across all baseline parameter fields and triggers the "Sign Up" submission event.
- **System Action:** Executes validation strings, commits the structural array data row to the database, logs in the user instance, and opens up the User Dashboard workspace immediately.

4.1.3 Functional Requirements

- **REQ-1 (Parameter Ingestion Matrix):** The registration sheet must capture specific string parameters for: Name, Email, Password, NID No, Phone Number, Age, and Gender.
- **REQ-2 (Gender Demographics):** The gender capture container must supply separate radio-type selector buttons explicitly for *Male*, *Female*, and *Other*.
- **REQ-3 (Credential Rule Verification):** The system must block form submissions unless the password field contains 5 characters and the phone array contains 11 numerical characters.
- **REQ-4 (Instant Portal Routing):** Upon successful submission of valid registration parameters, the application routing engine must bypass all intermediate landing pages and directly **open the User Dashboard**.

4.2 Old User Login Form

4.2.1 Description and Priority

Verifies credentials for returning registered profiles to establish active authenticated system states. (Priority: High)

4.2.2 Functional Requirements

- **REQ-5 (String Authentication):** The gateway must pull and cross-reference user-supplied inputs against matching Email and Password fields in the active user tables.
- **REQ-6 (Fault Exception Handling):** In the event of password verification mismatch, the interface must reject the login handshake and overlay a structural alert label reading: *"Invalid for your password"* or *"Forgot for your password?"*.
- **REQ-7 (Instant Dashboard Access):** Upon successful submission of confirmed user parameters during login, the system must immediately authenticate the background session state and **open the User Dashboard**.

4.3 Subscribed User Dashboard Console

4.3.1 Description and Priority

Aggregates and exposes therapeutic progress logs, upcoming schedules, and subscription-gated material. (Priority: High)

4.3.2 Functional Requirements

- **REQ-8 (Upcoming Appointments Queue):** The screen must display an expandable dashboard element labeled **Upcoming Appointments** showcasing an absolute running counter alongside active metadata (assigned practitioner name, targeted timestamp, and confirmation status).
- **REQ-9 (Completed Appointments Log):** The interface must house a structural card component tracking verified historic sessional entries labeled **Completed Appointments**.
- **REQ-10 (Daily Meditation Time Tracker):** The system must generate a dynamic multi-bar graph layer plotting total logged daily meditation minutes against a running calendar week.
- **REQ-11 (Gated Media Streaming Architecture):** Access to mental wellbeing **Vlog list directories**, text/article **Content libraries**, and premium **Magazine lists** must be conditionally locked behind a profile subscription validation check.

4.4 Doctor Enrollment System

4.4.1 Description and Priority

Processes application metadata and supporting physical evidence submitted by medical professionals for admin justification review. (Priority: High)

4.4.2 Functional Requirements

- **REQ-12 (Professional Metadata Collection):** The onboarding screen must render distinct parameters capturing: Full Name, Email, Password, Phone Number, Age, Gender, NID Number, ETIN ID, Experience level (constrained to a 0 to 5 year slider), Category, and a Professional Photo asset.
- **REQ-13 (Classification Enums):** The Category field model must implement exact checkbox entities for *Psychologist* and *Psychiatrist* segregation.
- **REQ-14 (Academic Matrix Checklist):** The credentials field must map professional designations via targeted Boolean checkboxes for: *PhD in Psychology, PsyD, MPhil Clinical Psychology, MSc/MA Clinical Psychology, BSc/BA Psychology, DM Psychiatry, MD Psychiatry, FCPS Psychiatry, DNB Psychiatry,* and *MBBS*.
- **REQ-15 (Binary Asset Ingestion Nodes):** The user interface layout must incorporate strict file attachment triggers ensuring the upload of: *NID National ID Copy*, *Medical Internship Certificate*, and a *Curriculum Vitae (CV)* document.
- **REQ-16 (Admin Justification Workflow):** Upon submission, the record status must flag automatically as Pending. The profile is restricted from the public Directory layout until an Administrator validates the uploaded documents and transforms the state to Justified.
- **REQ-17 (Instant Form Routing):** Upon successful submission of the entry components, the workflow routing module must immediately execute a window change and **open the Doctors Dashboard** console framework (holding data fields in their native unverified/pending state).

4.5 Doctor Dashboard Framework

4.5.1 Description and Priority

Supplies active clinical practice managers with real-time operational calendar views and patient metrics. (Priority: High)

4.5.2 Functional Requirements

- **REQ-18 (Daily Operational Load Metric):** The primary view grid must render a distinct quantitative status box mapping **how many patients need to be seen today**, pulling current row tallies from a block labeled *Today's Appointments*.
- **REQ-19 (Active Schedule Queue):** The central dashboard workspace must plot a timeline card containing chronological index blocks for all confirmed client sessions scheduled for the current date.
- **REQ-20 (Auxiliary Schedule Integration):** The layout must maintain an active sub-panel called *My Calendar* enabling doctors to synchronize external appointments, specify hourly shift allocations, and lock out unavailability blocks.

4.6 Fact-Based Screening Questionnaire Engine

4.6.1 Description and Priority

Evaluates psychological wellness metrics through localized multi-point fact calculations. (Priority: Medium)

4.6.2 Functional Requirements

REQ-21 (Fact-Based Assessment Items): The assessment questionnaire layer must present 10 uniform clinical symptom check prompts to the user.

REQ-22 (Scale Weights): Every question block must map to a standardized radio choice array comprising 5 specific value parameters: *Strongly Disagree (1 point), Disagree (2 points), Neutral (3 points), Agree (4 points), and Strongly Agree (5 points)*.

REQ-23 (Algorithmic Summation & Scoring Logic): Upon processing submission, the computing engine must calculate the total raw points (Scale: 10 to 50) and output a clean diagnostic evaluation card matching these exact code conditional blocks:

- **Score 10 – 19 (Severely Depressed):**
  - **Result Text Output:** "⚠️ Severely Depressed — Your responses indicate severe emotional distress. Immediate professional consultation with a doctor or mental health expert is strongly advised. Please do not hesitate to reach out for support."
- **Score 20 – 29 (Slightly Depressed):**
  - **Result Text Output:** "😟 Slightly Depressed — Mild symptoms detected. Your mental wellbeing is currently under strain. Consider practicing consistent stress-management, mindfulness, and dedicated self-care."
- **Score 30 – 39 (Totally Well):**
  - **Result Text Output:** "🌿 You Are Totally Well — Your mental wellbeing is balanced, resilient, and in a very healthy state. Keep maintaining this positive lifestyle!"
- **Score 40 – 50 (Exceptional Wellbeing):**
  Result Text Output: "🔥 Exceptional Wellbeing — You are the ultimate definition of mental wellness! You are beautifully balanced, highly resilient, and deeply inspired. Continue to shine and inspire others around you."

5. Other Nonfunctional Requirements

5.1 Performance Requirements

All user password arrays must be dynamically converted using a cryptographically sound salt hashing standard (BCrypt) prior to persistence operations.
Clinician identification documentation (NID uploads, license documents) must be encrypted at rest and isolated within an authorization layer restricted to authorized administrators.

5.2 Safety Requirements

The system must ensure that no harm, loss, or damage results from its use, especially since it deals with sensitive mental health data and wellbeing assessments.

Potential Risks:

- **Data Loss or Breach:** Unauthorized access to patient records, doctor credentials, or assessment results.
- **Misinterpretation of Results:** Users may misread wellbeing scores and delay seeking professional help.
- **System Downtime:** Service unavailability could prevent users from accessing urgent support.
- **Unauthorized Access:** Weak authentication could allow malicious actors to impersonate users or doctors.

5.3 Security Requirements

The system must ensure the confidentiality, integrity, and availability of all sensitive data related to users, doctors, and administrators. Given the nature of mental health information, strict security and privacy measures are mandatory.

User Identity Authentication:

- The system shall require secure login credentials (email + password) for all users.
- Passwords must be hashed using password_hash() and never stored in plain text.
- Duplicate email registrations must be prevented.
- Optional **Two-Factor Authentication (2FA)** may be implemented for enhanced security.
- Session management must ensure automatic logout after inactivity and prevent session hijacking.

Data Protection:

- All communication between client and server must use **HTTPS** encryption.
- Sensitive data (user profiles, assessment scores, doctor credentials) must be encrypted at rest in the database.
- Role-based access control must be enforced (patients, doctors, admins) to prevent unauthorized access.
- Audit logs must be maintained for login attempts, data changes, and administrative actions.

Privacy Safeguards:

- Patient data must not be shared with third parties without explicit consent.
- Doctors' enrollment documents must be securely stored and accessible only to authorized administrators.
- Chatbot interactions must be anonymized and stored securely to protect user privacy.
- Error messages must be user-friendly and must not expose system internals or database queries

5.4 Software Quality Attributes

The following quality characteristics are critical for the success of the Mental Wellbeing System. They are defined to ensure the product meets both customer expectations and developer standards.

Adaptability:

- The system shall support future integration with tele-consultation modules (video calls) without major redesign.
- The chatbot module shall be adaptable to new AI APIs or NLP engines.

Availability:

- The system shall maintain **99.5% uptime** to ensure users can access mental wellbeing services reliably.
- Scheduled maintenance windows shall be communicated at least 24 hours in advance.

Correctness:

- Assessment scoring must strictly follow predefined logic (e.g., score ranges 10–19, 20–29, etc.) with no deviation.
- All calculations and displayed results must be verified against test cases before deployment.

Flexibility:

- The system shall allow customization of questionnaires and scoring rules by administrators.
- Subscription models (monthly, yearly) can be modified without affecting core functionality.

Interoperability:

- The system shall integrate with external APIs (payment gateway, email service, chatbot engine) using standard REST protocols.
- Data exchange shall use **JSON** format with UTF-8 encoding.

Maintainability:

- Code shall follow modular design principles to allow easy updates.
- Documentation shall be maintained for all modules to support future developers.

Portability:

- The system shall run on any modern browser (Chrome, Edge, Firefox, Safari).
- Deployment shall be possible on both on-premise servers and cloud platforms (AWS, Azure).

Reliability:

- The system shall recover gracefully from unexpected failures (e.g., database disconnection).
- Automated backups shall ensure no data loss beyond 24 hours.

Reusability:

- Authentication and session management modules shall be reusable across other projects.
- UI components (buttons, forms, cards) shall follow a consistent design library.

Robustness:

- The system shall handle invalid inputs gracefully, displaying clear error messages without crashing.
- Security safeguards shall prevent SQL injection, XSS, and CSRF attacks.

Testability:

- Each module shall include unit tests with at least **80% code coverage**.
- Integration testing shall verify communication between frontend and backend APIs.

5.5 Business Rules

The following operating principles define how different roles and individuals may interact with the system under specific circumstances. These rules are not functional requirements themselves but imply certain requirements to enforce them.

User Rules (Patients):

- Patients must register and log in before accessing assessment tools, chatbot, or appointment booking.
- A patient can only book appointments with verified doctors.
- Patients may view their own assessment results and subscription status but cannot access other users' data.

- Patients must provide consent before sharing assessment results with doctors.

Doctor Rules:

- Doctors must complete enrollment and submit required credentials before gaining access to the system.
- Only verified and approved doctors can view patient appointment requests.
- Doctors may update their availability schedule but cannot modify patient assessment data.
- Doctors must comply with medical board regulations when providing consultations.

Administrator Rules:

- Administrators are responsible for verifying doctor credentials and approving enrollment.
- Administrators may manage subscription plans, questionnaires, and system configurations.
- Administrators can suspend or deactivate accounts that violate system policies.
- Administrators must ensure compliance with data protection regulations (GDPR, HIPAA).

System Rules:

- Duplicate email registrations are not allowed.
- Passwords must be stored securely using hashing algorithms.
- Assessment scoring must follow predefined ranges (10–19, 20–29, 30–39, 40–50).
- The system must enforce role-based access control (patients, doctors, admins).
- All sensitive data must be encrypted in transit and at rest.